# Miles Zelasko

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## Education

**University of New Hampshire** — Durham, NH
*Bachelor of Science in Mechanical Engineering | GPA: 3.45/4.00* — *Expected May 2027*

## Technical Skills

**Languages**: Python, MATLAB, Julia, Lua/Luau
**Software & Tools**: Git, VS Code, SolidWorks
**Libraries**: pandas, NumPy, Matplotlib, CoolProp, DataFrames.jl, DifferentialEquations.jl, Measurements.jl, Statistics.jl
**Certifications**: CSWA (Certified SolidWorks Associate)

## Projects

**Tuned Liquid Damper for Structural Sway Mitigation** | *MATLAB, Julia, Arduino* — March 2026 – May 2026

- Designed and tested a 3D-printed U-tube liquid damper on a scaled cantilever building model, instrumented with an MPU-6050 accelerometer and Arduino, sizing the water volume from the measured natural frequency
- Extracted natural frequency and damping ratio in MATLAB using PCA and peak-decay fitting: the damper more than doubled damping ($\zeta = 0.0093$ to $0.0193$, $+108\%$), while an equal-mass static weight reduced it by 16%, indicating sloshing rather than added mass as the cause

**Data Center Thermal Management & Cooling System Design** | *MATLAB, CoolProp* — December 2025

- Developed parametric Rankine cycle and vapor-compression refrigeration models, evaluating turbine inlet conditions and condenser temperature effects on thermal efficiency and COP
- Selected optimal facility location via climate data analysis across four ASHRAE regions, establishing thermal boundary conditions for the integrated system design

**Rock Climbing Training Tool** | *SolidWorks* — June 2024 – June 2025

- Designed a finger strength training tool in SolidWorks, iterating based on feedback across beginner to advanced climbers to refine load and grip geometry
- Reduced weight >50% across prototype iterations, which cut manufacturer-quoted cost 43%; prepared CAD for outsourced fabrication in 6061 aluminum and specified surface finish

## Experience

**Senior Sales Associate** — June 2021 – August 2024
*Portland Discovery Land & Sea Tours* — *Portland, ME*

- Managed end-to-end boarding logistics for 500+ daily customers, coordinating across multiple vessel and trolley operations simultaneously
- Trained new team members on point-of-sale systems, policies, and operational procedures, reducing onboarding time
- Oversaw opening and closing procedures, ensuring consistent adherence to operational standards
- Supported back-office operations including administrative and internal communications

## Extracurriculars

**RL$^2$** — September 2025–Present
*Member* — *University of New Hampshire*

- Participated in graduate-level reading groups on reinforcement learning theory led by UNH faculty
- Engaged with research topics including robust MDPs, constrained optimization, and average reward criteria